🧠 Scenario-Based MCQs: Microsoft Fabric (Enterprise Level)

🏢 Scenario 1: Unified Data Platform Migration

Q1

The team wants a single storage layer for structured and unstructured data with ACID transactions.

A. Data Warehouse

B. Lakehouse

C. Power BI Dataset

D. Dataflow Gen2

Answer = B. Lakehouse

Q2

Data engineers need to process large-scale data using Spark while maintaining SQL accessibility.

A. Data Warehouse

B. Lakehouse with Delta tables

C. Power BI

D. Data Activator

Answer = B. Lakehouse with Delta tables

Q3

Business users want to build reports directly on top of the unified data without duplication.

A. Export data to Excel

B. DirectLake mode

C. Import mode only

D. CSV extracts

Answer = B. DirectLake mode

🔄 Scenario 2: Data Ingestion & Pipeline Orchestration

Q4

They need a low-code ingestion tool for transforming and loading data.

A. Notebook

B. Dataflow Gen2

C. SQL endpoint

D. Power BI Dashboard

Answer = B. Dataflow Gen2

Q5

The pipeline must orchestrate multiple steps including ingestion, transformation, and model refresh.

A. Manual execution

B. Fabric Data Pipelines

C. Excel macros

D. Power BI visuals

Answer = B. Fabric Data Pipelines

Q6

A pipeline fails intermittently due to API latency.

A. Ignore failures

B. Implement retry policies and logging

C. Reduce data size

D. Run pipeline once

Answer = B. Implement retry policies and logging

🧱 Scenario 3: Lakehouse Architecture & Delta Tables

Q7

A team wants version control and time travel capabilities on their data.

A. CSV files

B. Delta Lake tables

C. Excel storage

D. JSON files

Answer = B. Delta Lake tables

Q8

Data updates need to support merge (insert/update/delete) operations efficiently.

A. Append-only data

B. Delta Lake MERGE

C. Recreate tables daily

D. Use Excel

Answer = B. Delta Lake MERGE

Q9

Which layer in Medallion architecture stores raw ingested data?

A. Gold

B. Silver

C. Bronze

D. Semantic

Answer = C. Bronze

📊 Scenario 4: Data Warehousing & SQL Analytics

Q10

A finance team requires high-performance SQL queries on structured data.

A. Lakehouse only

B. Fabric Data Warehouse

C. Power BI only

D. Dataflow

Answer = B. Fabric Data Warehouse

Q11

The team wants separation of compute and storage for scalability.

A. Dataflows

B. Warehouse engine

C. Excel

D. Dashboards

Answer = B. Warehouse engine

Q12

Complex joins and aggregations are slow in Lakehouse SQL endpoint.

A. Use CSV

B. Move workload to Data Warehouse

C. Reduce joins

D. Use Power BI

Answer = B. Move workload to Data Warehouse

📈 Scenario 5: Power BI Integration

Q13

A report needs real-time data access without scheduled refresh.

A. Import

B. DirectLake

C. Excel export

D. Snapshot

Answer = B. DirectLake

Q14

Users complain about slow dashboards due to large datasets.

A. Increase visuals

B. Use aggregations and semantic model optimization

C. Add more columns

D. Export data

Answer = B. Use aggregations and semantic model optimization

Q15

A single version of truth is required across reports.

A. Multiple datasets

B. Centralized semantic model

C. Excel sheets

D. CSV sharing

Answer = B. Centralized semantic model

🔐 Scenario 6: Governance & Security

Q16

Sensitive financial data must be restricted based on user roles.

A. Dataflows

B. Row-Level Security (RLS)

C. Pipelines

D. Notebooks

Answer = B. Row-Level Security (RLS)

Q17

The organization wants to track data lineage and dependencies.

A. Dashboards

B. Lineage view in Fabric

C. Excel logs

D. Manual tracking

Answer = B. Lineage view in Fabric

Q18

Data access must comply with enterprise security policies across all workloads.

A. Manual permissions

B. Centralized governance in Fabric workspace

C. Share links

D. Export data

Answer = B. Centralized governance in Fabric workspace

⚙️ Scenario 7: Real-Time Analytics & Monitoring

Q19

A company wants to trigger alerts when sales drop below threshold in real-time.

A. Power BI import mode

B. Data Activator

C. Excel formulas

D. Dataflows

Answer = B. Data Activator

Q20

Streaming data from IoT devices needs to be processed and visualized instantly.

A. Batch processing

B. Real-Time Analytics in Fabric

C. CSV storage

D. Manual refresh

Answer = B. Real-Time Analytics in Fabric